Anderson Cicero Sena Freitas

Head of Engineering | AI Platform & Backend | Python · Django · FastAPI · Go · React · AWS

Personal Details

Palmas, TO, Brazil
Phone: +55 63 99204-1379
freitas.dev@proton.me LinkedIn| GitHub | Portfolio

Software Engineer working across backend, AI platforms and healthcare systems, building scalable, secure solutions focused on real business impact.

Solid experience with Python (Django/FastAPI), Go, PostgreSQL, Redis and asynchronous processing (Celery/RabbitMQ), building APIs and decoupled architectures. Hands-on with AWS, Docker, Terraform and CI/CD pipelines, including production incident response.

I currently lead engineering for an enterprise multi-tenant AI agent platform (LLM orchestration, MCP, RAG and Microsoft 365 integrations) and work on healthcare products and data engineering (Airflow, Superset).

Experience in government and private-sector projects, including critical public healthcare systems and data platforms, with a focus on scalability, governance and performance.

Technical Skills

Artificial Intelligence & Data

- LLMs (OpenAI, Anthropic, OpenRouter, LiteLLM, Ollama)
- AI agents and multi-agent orchestration
- MCP (Model Context Protocol)
- RAG and vector search (PGVector)
- Data engineering: Airflow, Apache Superset

Backend

- Python (Django, DRF, Django Ninja, FastAPI)
- Go (Fiber)
- Node.js (Express)
- PostgreSQL, MariaDB, MongoDB, Redis
- Celery, RabbitMQ, WebSockets (Django Channels)

- Microsoft Graph, Entra ID, OAuth2/JWT

Frontend

- React, Next.js
- Angular, Vue
- TypeScript
- Tailwind CSS, ShadCN, Storybook

DevOps & Architecture

- Docker, Nginx, Traefik, Caddy
- AWS (EC2, Lambda, S3, ECR, Aurora, SES) and Azure
- Terraform
- CI/CD (GitHub Actions, GitLab, Bitbucket Pipelines, Azure DevOps)
- Observability: Prometheus, Grafana, Loki

- REST APIs, SSE

Experience

Zuraio (Switzerland)

Head of Engineering
10/2026 - Present
Founding AI Platform Engineer
02/2026 - 10/2026

- Architected the multi-tenant AI platform across 15+ interoperable services: tenant isolation, authentication, agent orchestration, enterprise integrations and infrastructure.
- Go authentication service (Fiber + Redis) with rate limiting and Prometheus metrics, plus host-based multi-tenant routing with cache invalidation via Redis Pub/Sub.
- Multi-agent orchestration with an ethics gate and multiple execution strategies, plus the internal zuraio-mcp-core SDK for MCP-based agents.
- Microsoft 365 integration via Microsoft Graph and Entra ID: email agent with RAG over the mailbox, SharePoint document access and Office add-ins.
- Web research and ingestion pipeline with Crawl4AI, SearXNG and a Django control plane with async queues (arq/Redis), feeding the knowledge base with OCR, chunking and vector search.
- Containerized infrastructure (development, staging and production) with GitHub Actions, GHCR, security scanning (Trivy) and observability with Prometheus, Grafana and Loki.
- @zuraio/ui design system (React 19, ShadCN, Storybook) adopted across 4 frontends, with internationalization in 4 languages.

Medflow Tech
Senior Software Engineer
03/2026 - Present

- Evolved the clinical protocol platform (Django and React): protocol execution embedded in partner portals, referrals and medical guidance, and consumption of FHIR resources.
- Backend of a mental health care coordination system (Django 5.2, DRF/Ninja, Celery/RabbitMQ, WebSockets), with role-based access control, audit logging, S3/Azure Blob attachments and PDF generation of individual care plans.
- Migrated the protocol improvement hub from Supabase to a dedicated Django API on AWS (EC2, S3, Terraform, Bitbucket Pipelines), with a status state machine, a tested permission matrix and httpOnly cookie authentication.
- Airflow data pipelines (partner API ingestion, Aurora PostgreSQL views, SES alerts) and migration of a Power BI dashboard to Apache Superset managed as code.

State Department of Health of Tocantins (Brazil)
Software Engineer
07/2025 - 03/2026

- **Integration Bus for the State Health Data Network (REDS-TO)**
    - Technical owner of the architecture and development of the REDS data bus.
    - Event-driven distributed architecture (FastAPI, Go), PostgreSQL, asynchronous processing and observability
    - Security, governance and compliance in a critical public healthcare environment

- **Portal CIEGES**
  - Technical owner of the portal for the Strategic Intelligence Center for State SUS Management of Tocantins
  - Backend with Django REST Framework, React frontend and PostgreSQL, GitLab CI/CD, automated tests and a focus on scalability

Prova Fácil - Conecta Ecosystem
Junior Full Stack Developer
02/2025 - 07/2025

- Took part in production incident war rooms on the main product (a Django monolith), diagnosing and shipping fixes alongside the Tech Lead, often owning a workstream independently.
- Developed and maintained features in the Django monolith and in serverless functions on AWS (Lambda, Python).
- Asynchronous processing with RabbitMQ and Celery in Docker environments integrated with AWS.
- Frontend with Vue.js, Angular and React.

Institute for Urban Care
Developer
06/2024 - 07/2025

- Developed and maintained institutional systems and applications supporting public management.
- Worked with Kotlin, Spring Boot, PostgreSQL, Angular and React Native.
- Took part in data analysis and technical requirements definition.

Digital Solutions Office (Protic) - Federal University of Tocantins
Developer
08/2022 - 03/2025

- Developed and maintained systems with Python/Django (DRF), JavaScript and MariaDB, REST APIs, SSE, data modeling and requirements gathering.

CECANE - Federal University of Tocantins
Developer
03/2022 - 03/2024

- Developed and maintained systems with Python/Django (DRF), MariaDB and REST APIs supporting family farming.

Some of these roles ran in parallel across academic, institutional and separate contracts.

Technical Projects

ArmoredDjango

Production-ready Django template focused on security and sound architecture practices. JWT authentication, custom user model, strong password validation, rate limiting, email system, caching and automated tests.
Architecture ready for CI/CD and Docker environments; used as the API foundation in production at Medflow (protocol improvement hub).

Technologies: Python, Django, Django REST Framework, PostgreSQL, Docker, Pytest, GitHub Actions.

Noeris

Intelligent platform for data visualization and decision support, with controlled LLM integration, cognitive question classification and an architecture that lets AI models evolve independently.
Built a classification pipeline and natural-language insight generation, with a clear split between frontend, backend and the AI layer, ensuring scalability and testability.

Technologies: React, TypeScript, Vite, Tailwind CSS, Django, Django REST Framework, PostgreSQL, Docker, classification models, LLMs.

PlantAgora

Web and mobile platform for community garden management, with user management, geospatial data and REST APIs.
Social-impact application with a full-stack architecture.

Technologies: Django, React Native, PostgreSQL, REST API.

Education

Bachelor of Computer Science

Federal University of Tocantins 2021 – In progress

Courses and Additional Training

Python Backend & APIs

- Building REST and asynchronous APIs with FastAPI
- Authentication, automated testing and API deployment
- Full Stack Development with Django
- Python best practices and relational databases

Frontend

- Building interfaces with React and TypeScript
- Angular fundamentals and service-based architecture

DevOps and Software Engineering

- Docker, CI/CD and deployment automation
- Version control with Git and GitHub
- REST API fundamentals, HTTP and service integration
- Agile Methodologies (Scrum / Kanban)

Languages

Portuguese: Native
English: Intermediate (technical reading and professional communication)